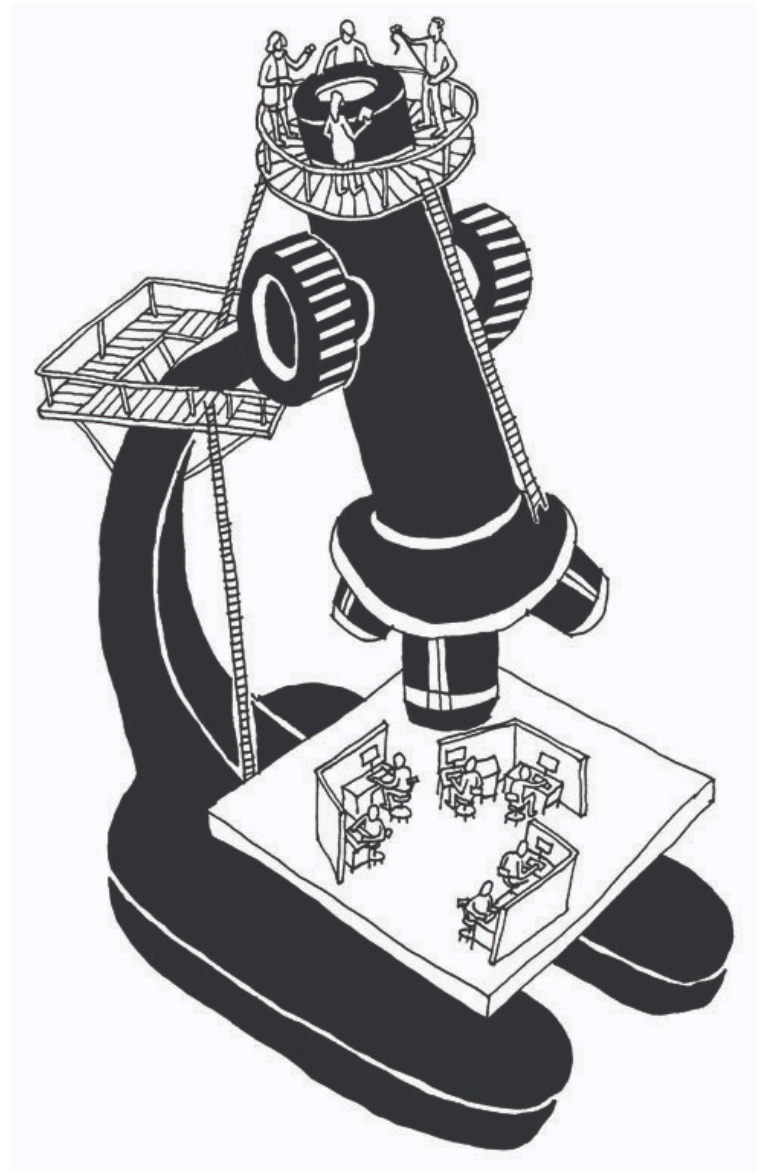


ECS 260: Software Engineering



Who We Are

- Instructor: Prof. Filkov
 - PhD 2002
 - Full Prof in CS
 - Specialty: AI/Data Science in SE, AI in Health (especially as relating to social matters in SE)
- TA: Raiyan Jahangir

Who are you?

- #PhD:
- #MS:
- #non-CS:
- #non-coders:
- #Systems (SE/PL, OS, Software, Networking?)

This Class

- Advanced Software Engineering class, geared toward applied research
- **Empirical Software Engineering:** studying Software Engineering topics from an empirical science perspective
- Lectures:
 - Theory: science, methods
 - Practice: how to do it (papers, tools, best practices, etc)

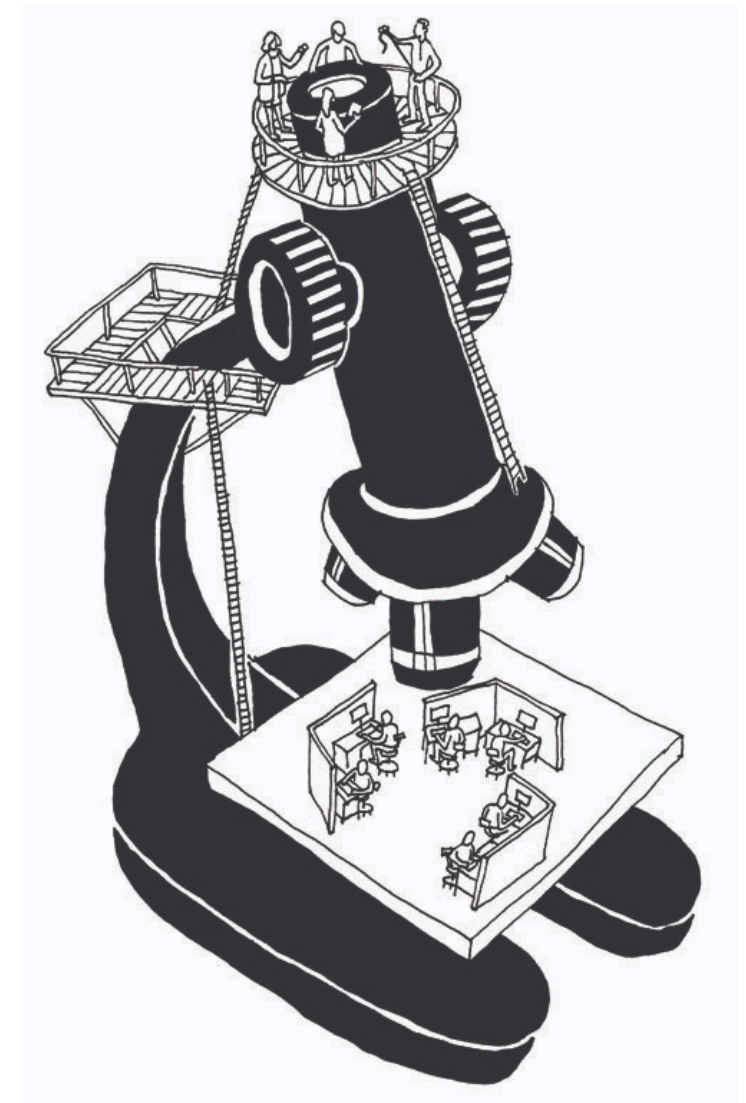


Illustration by Tom Dunne.

Workload

- **Weekly reading:** 2-4 papers, chapters, slides
- **Quizzes:** weekly, see schedule
- **Group presentation:** due Jan 28, papers review (to establish background for your project)
- **Projects:** Team work, resulting in solid preliminary results. Deliverables: proposal, progress, and final report. All will be graded

Projects

- **Team-up** by Jan 11! 4-5 people per team. We can help.
- **Project proposal** due Jan 16. See list of projects on canvas (TOP SECRET!)
 - Check out the resources page for tools for data gathering and analysis.
- **Progress report:** midway through quarter, due Feb 18.
- **Check-in:** First week of March.
- **Final report**, due Mar 18. Must be written as a technical paper.

Lecture 1

- Empirical Science
- Science Methods

Reading: Chapter 1 from Research design: Qualitative, quantitative, and mixed methods approaches

Software Engineering and Empirical Software Engineering

- Nowadays, code is large. Software projects are large. Programming teams are large. The tools they use are many.
- Thus, modern software development is complex
 - a) many moving elements
 - b) non-linear dependencies
 - c) multiple outcomes and goals
- Yet, Software Engineering folks **have traditionally not been big on scientifically assessing their tools or rigorously validating their theories and assumptions**

What are some theories?
- Solution: ESE

Empirical Software Engineering

- **Empirical software engineering** is an applied research area concerned with the empirical observation of software engineering artifacts and the empirical validation of software engineering theories and assumptions.
- Subfields of software engineering that are accustomed to empirical research comprise **software evolution, software maintenance and mining software repositories**.
- The type of problems ESE is concerned with:
 - Observation of artifacts (code, bug reports, email discussions, etc)
 - Validation of tools (code quality of C vs JavaScript)
 - Validation of methodologies (agile vs regular programming)
 - Validation of assumptions (strongly typed languages are better)

Inspired by Martin Monperrus

Examples of ESE Research

- Are more sophisticated code complexity measures, like McCabe's cyclomatic complexity, better at predicting code defects than LOC count?
- Is pairs programming better (wrt speed, quality) than singles programming?
- What is the higher maintenance burden of AI assisted coding due to?
- What is the best model to predict effort needed to deliver new code based on specs?
- Does the architecture of the software produced correlate with its quality?
- Which LLM tool is the “best” coding assistant?

ESE Paper Outlets

- Conferences
 - ICSE: International Conference on Software Engineering
 - FSE: Foundations of Software Engineering
 - ASE: Automated Software Engineering (2024 in Sac!)
 - MSR: Mining Software Repositories
- Journals
 - ESE: Empirical Software Engineering
 - ACM TOSEM: Transactions of Software Engineering and Methods
 - IEEE TSE: Transactions of Software Engineering

Empirical Research in General

- **Observation based** (thus it is closer to science than engineering, more on that in a second)
 - What can you observe, measure, bin, quantify?
 - Data, data, data
- How do you reason about the observations?
 - Methods
- Where do the observations fit in the bigger scheme of understanding phenomena?
 - Must have theory



Engineering vs. Science

→ Traditional View:

Scientists...

- create knowledge
- study the world as it is
- are trained in scientific method
- use explicit knowledge
- are thinkers

Engineers...

- apply that knowledge
- seek to change the world
- are trained in engineering design
- use tacit knowledge
- are doers

→ More realistic View

Scientists...

- create knowledge
- are problem-driven
- seek to understand and explain
- design experiments to test theories
- prefer abstract knowledge
- but rely on tacit knowledge

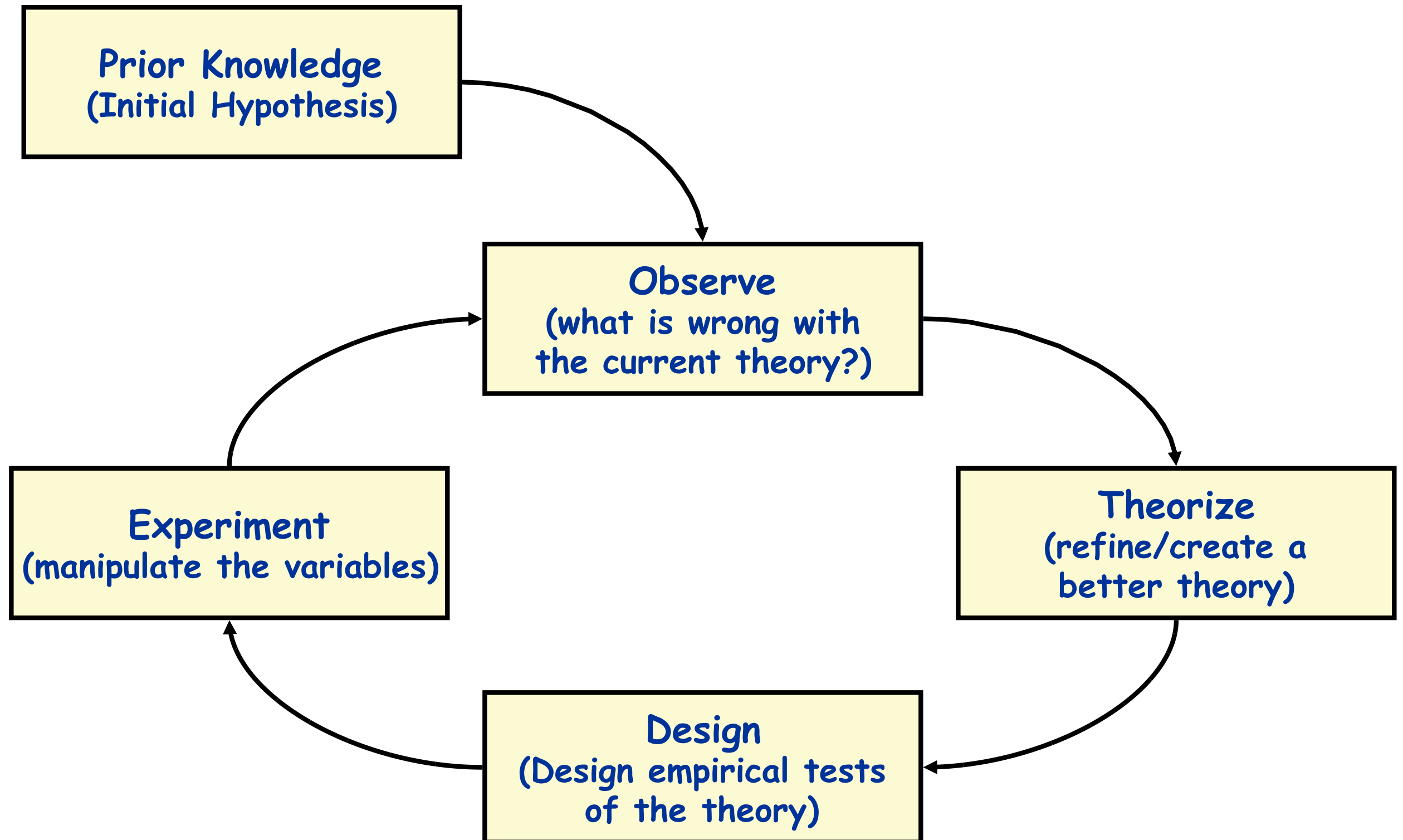
Engineers...

- create knowledge
- are problem-driven
- seek to understand and explain
- design devices to test theories
- prefer contingent knowledge
- but rely on tacit knowledge

Both involve a mix of design and discovery



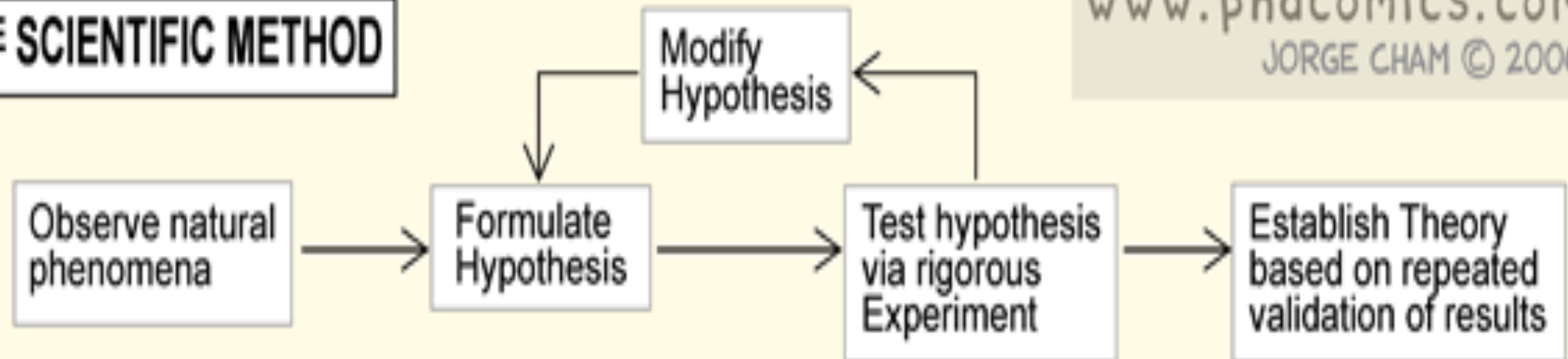
Scientific Inquiry



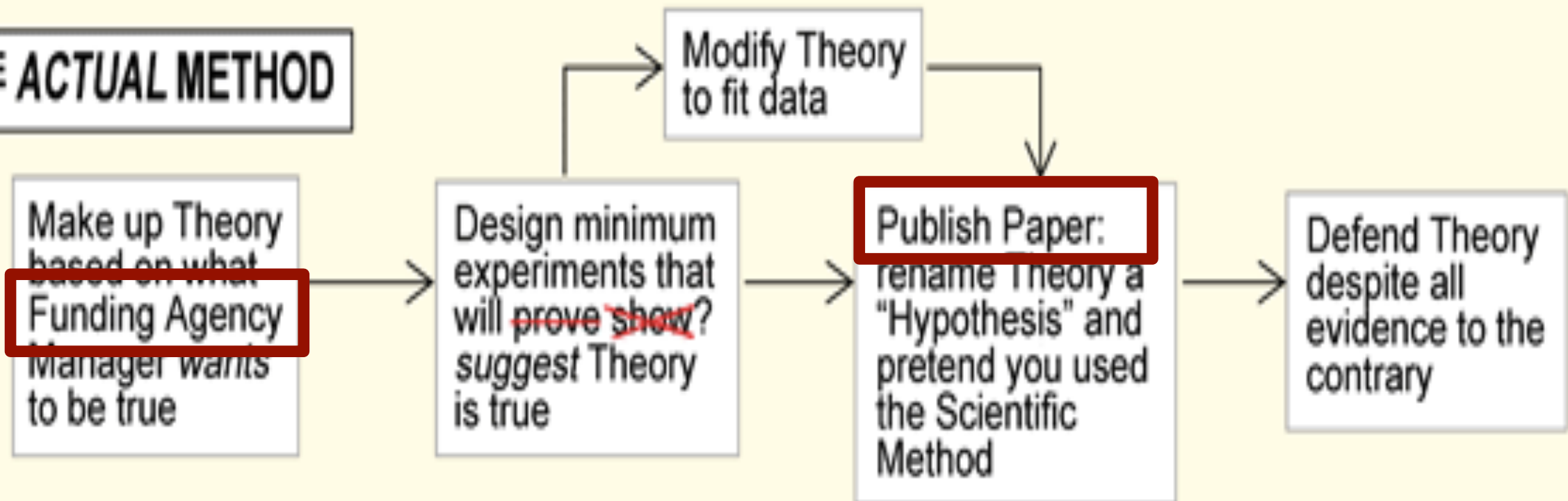


www.phdcomics.com
JORGE CHAM © 2006

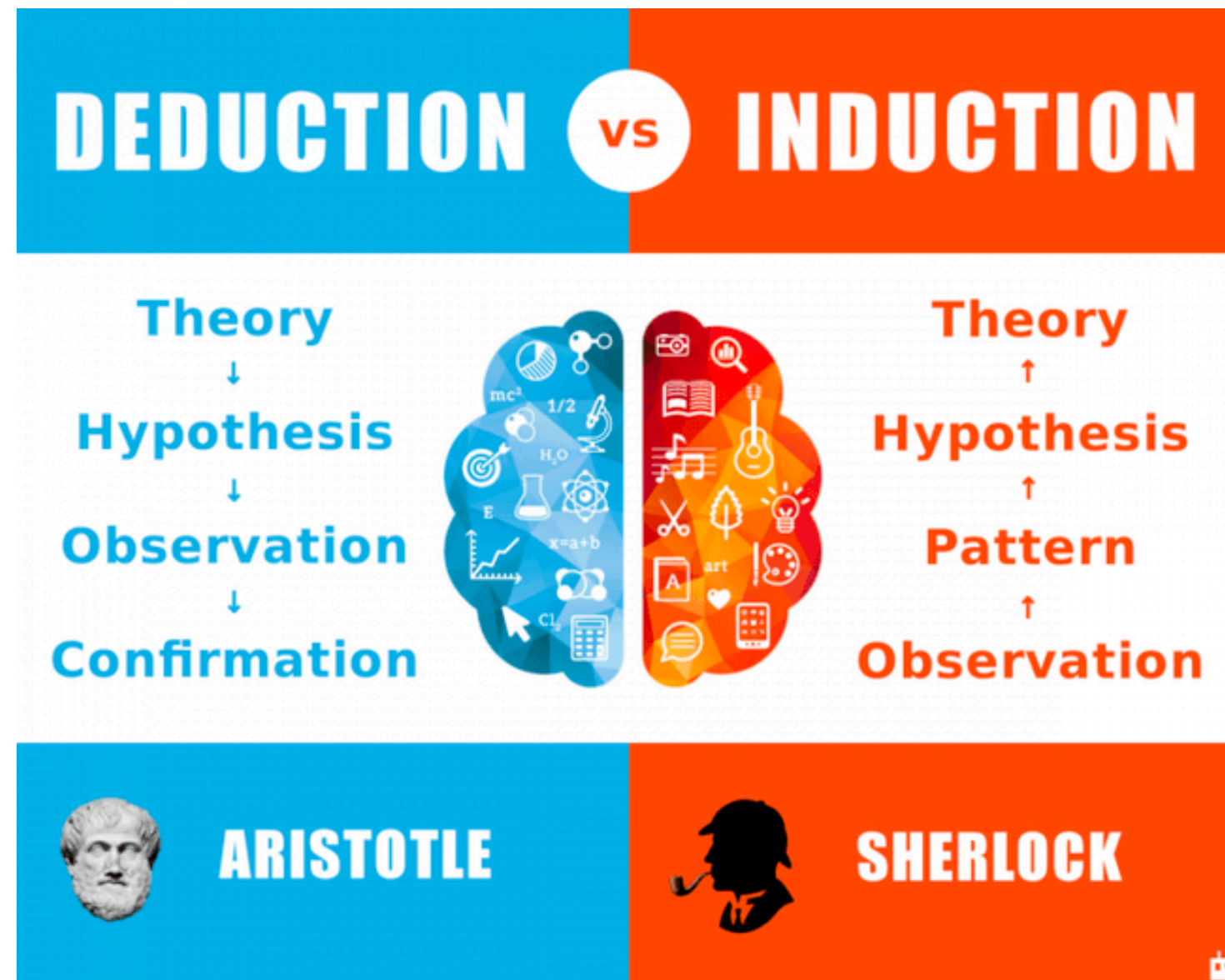
THE SCIENTIFIC METHOD



THE ACTUAL METHOD



Types of Inference/Reasoning



All men are mortal. Aristotle is a man. Therefore, Aristotle is mortal.

Deduction

All cars in this town drive on the right side of the street. Therefore, all cars in all towns probably drive on the right side of the street.

Induction

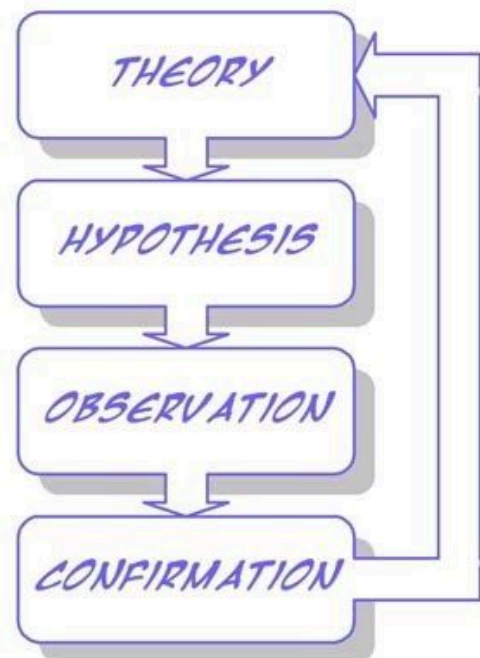
I have a bag of many coins. I've pulled 10 at random and they've all been pennies. Therefore, this is probably a bag full of pennies.

Induction

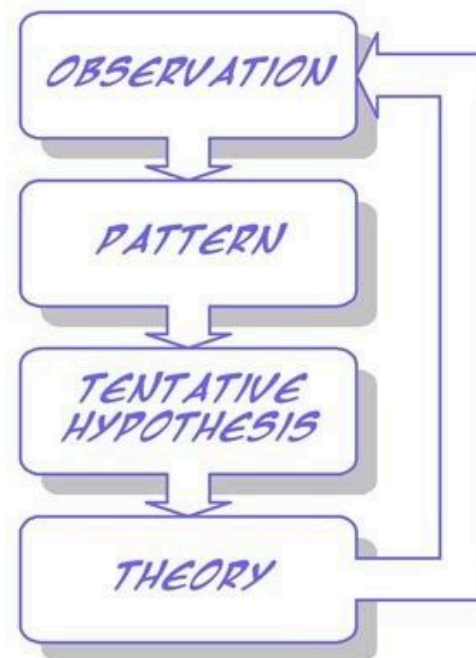
By DANIEL MIESSLER

Types of Inference/Reasoning

DEDUCTION



INDUCTION



By DANIEL MIESSLER

1. Deduction gets you to a perfect conclusion—but only if all your premises are 100% correct.
2. Deduction moves from theory to experiment to validation, where induction moves from observation to generalization to theory.
3. Deduction is harder to use outside of lab/science settings because it's often hard to find a set of fully agreed-upon facts to structure the argument.
4. Induction is used constantly because it's a great tool for everyday problems that deal with partial information about our world, and coming up with usable conclusions that may not be right in all cases.
5. Be willing to use both types of reasoning to solve problems, and know that they can often be used together cyclically as a pair, e.g., use induction to come up with a theory, and then use deduction to determine if it's actually true.
6. The main thing to avoid with these two is arguing with the force of deduction (guaranteed to be true) while actually using induction (probability based on strength of evidence).

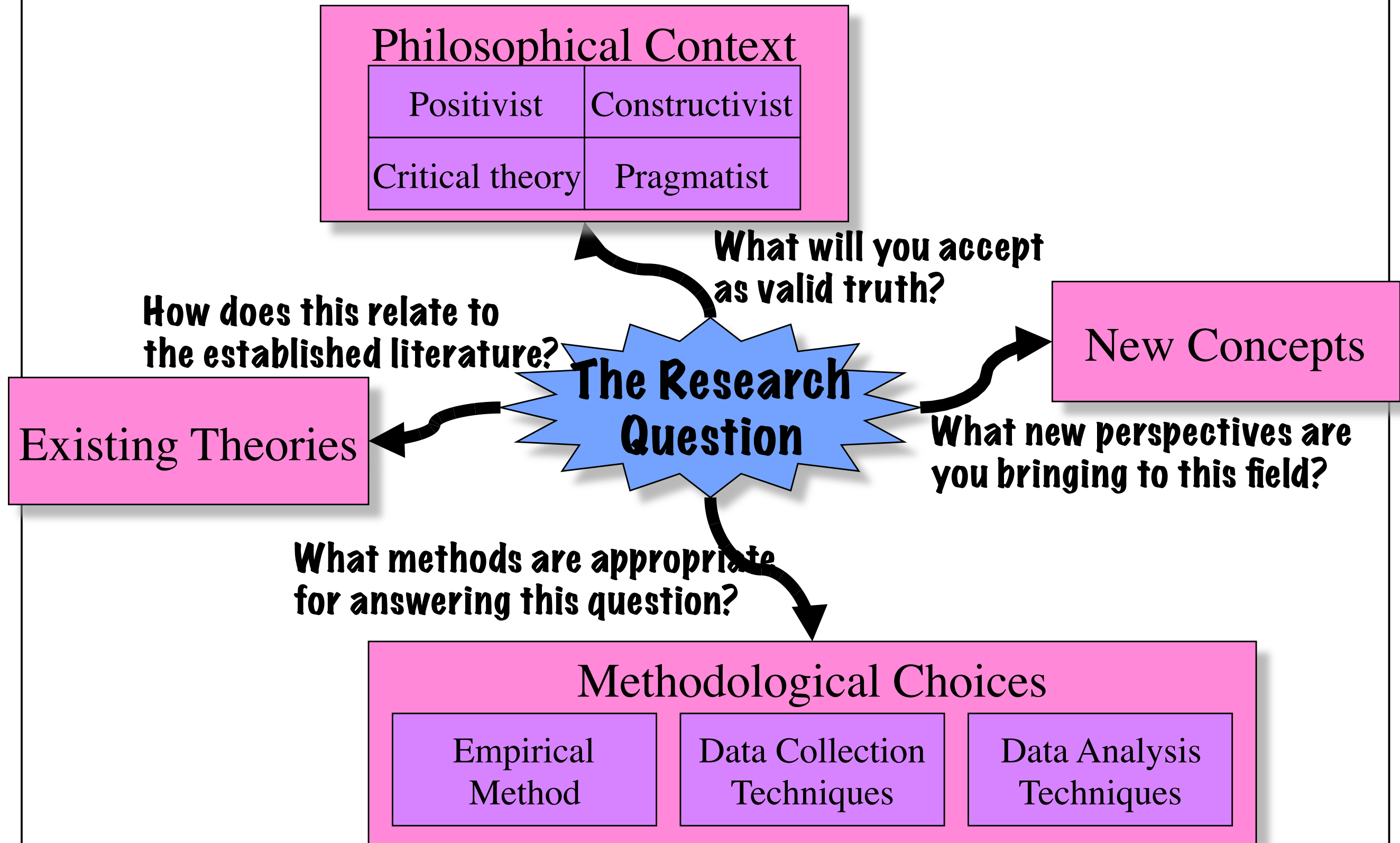
There is also Abductive Reasoning defined as reasoning that looks for the explanation that best fits the data. It is related to Inductive reasoning and it is appropriate when you can enumerate many possibilities (can explore solution space, e.g., AI)

Research Approaches

Chapter 1 - Creswell, J. W., & Creswell, J. D. (2017). Research design: Qualitative, quantitative, and mixed methods approaches. Sage publications.

- First step to design a research proposal: adopt a general (and guiding) framework
- Three frameworks (approaches to research):
QUALITATIVE, QUANTITATIVE, MIXED (qual + quant)
- These approaches include:
 - philosophical assumptions (postpositivist, constructivist, emancipatory/critical, pragmatic)
 - strategies of inquiry (experimental, ethnographic, narrative, mixed methods design)
 - research methods (data collection and analysis)

Putting the Question in Context



What will you accept as knowledge?

○ Positivist (or “Post-positivist”)

- Knowledge is objective
- “Causes determine effects/outcomes”
- Reductionist: study complex things by breaking down to simpler ones
- Prefer quantitative approaches
- **Verifying (or Falsifying) theories**

○ Constructivist/Interpretivist

- Knowledge is socially constructed
- Truth is relative to context
- Theoretical terms are open to interpretation
- Prefer qualitative approaches
- **Generating “local” theories**

○ Critical Theorist

- Research is a political act
- Knowledge is created to empower groups/individuals
- Choose what to research based on who it will help
- Prefer participatory approaches
- **Seeking change in society**

○ Eclectic/Pragmatist

- Research is problem-centered
- “All forms of inquiry are biased”
- Truth is what works at the time
- Prefer multiple methods / multiple perspectives
- **seeking practical solutions to problems**

What type of question are you asking?

- Existence:
 - Does X exist?
- Description & Classification
 - What is X like?
 - What are its properties?
 - How can it be categorized?
 - How can we measure it?
 - What are its components?
- Descriptive-Comparative
 - How does X differ from Y?
- Frequency and Distribution
 - How often does X occur?
 - What is an average amount of X?
- Descriptive-Process
 - How does X normally work?
 - By what process does X happen?
 - What are the steps as X evolves?
- Relationship
 - Are X and Y related?
 - Do occurrences of X correlate with occurrences of Y?
- Causality
 - Does X cause Y?
 - Does X prevent Y?
 - What causes X?
 - What effect does X have on Y?
- Causality-Comparative
 - Does X cause more Y than does Z?
 - Is X better at preventing Y than is Z?
 - Does X cause more Y than does Z under one condition but not others?
- Design
 - What is an effective way to achieve X?
 - How can we improve X?

What type of question are you asking?

- Existence:
 - Does X exist?
- Description & Classification
 - What is X like?
 - What are its properties?
 - How can it be categorized?
 - How can we measure it?
 - What are its components?
- Descriptive-Comparative
 - How does X differ from Y?

- Frequency and Distribution
 - How often does X occur?
 - What is an average amount of X?
- Descriptive Process
 - How does X normally work?
 - By what process does X happen?
 - What are the steps as X evolves?

- Relationship
 - Are X and Y related?
 - Do occurrences of X correlate with occurrences of Y?

- Causality
 - Does X cause Y?
 - Does X prevent Y?
 - What causes X?
 - What effect does X have on Y?
- Causality-Comparative
 - Does X cause more Y than does Z?
 - Is X better at preventing Y than is Z?
 - Does X cause more Y than does Z under one condition but not others?

- Design
 - What is an effective way to achieve X?
 - How can we improve X?



Many available methods...

Common “in the lab” Methods

- Controlled Experiments
 - Rational Reconstructions
 - Exemplars
 - Benchmarks
 - Simulations

Common “in the wild” Methods

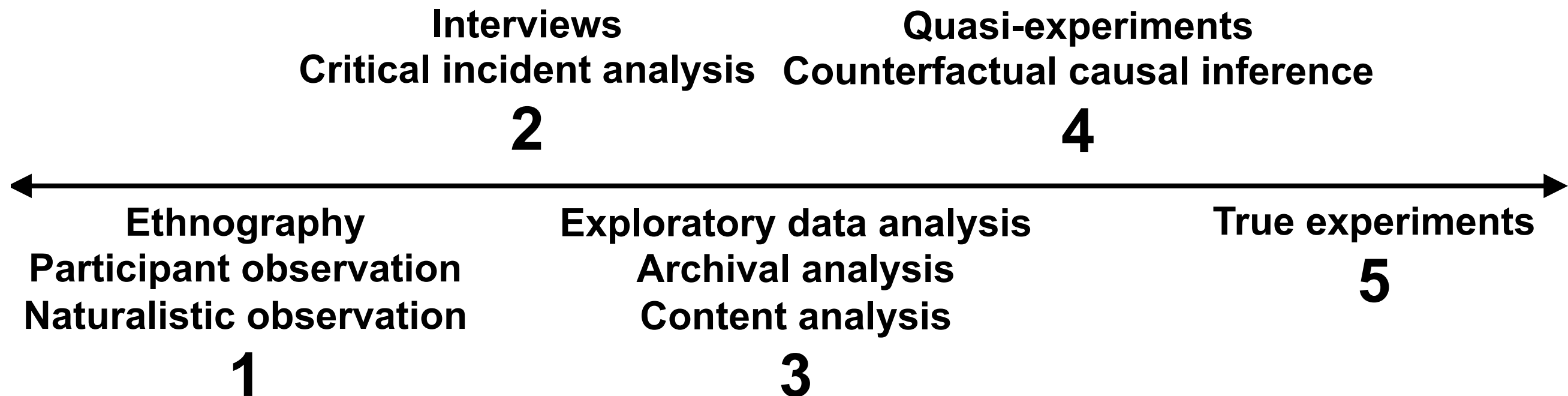
- Quasi-Experiments
- Case Studies
- Survey Research
- Ethnographies
- Action Research

○ Artifact/Archive Analysis (“mining”!)

One Way of Thinking About How Methods Are Related

- Roughly speaking, methods form a dimension
- Often most effective to use in combination or in sequence (rich to precise)
- Poles
 - Very little knowledge: “What’s going on?”
 - Extensive knowledge: “Precisely how is this different?”
- Generally, from qualitative to quantitative
 - From observation to experimentation
 - From richness to precision
 - From reliance on human interpretation to reliance on decision rules

Points on Empirical Methods Dimension



Example: Ethnography Rich precise

- What is it?
 - Immersion in the environment, group
 - Attempt to see the world through their eyes
- What questions can it answer?
 - How do the participants think about their work?
 - What are the problems?
- What makes it rigorous?
 - Constantly testing interpretations
 - Triangulation – multiple sources of data
- Example study?
 - Grinter: software architects

Example: Interviews Rich precise

- What is it?
 - Structured interaction
 - Questions, answers, followup
- What questions can it answer?
 - Perceptions, opinions, processed observations
 - How things are done, exceptions, problems
- What makes it rigorous?
 - Preparation with well thought-out topics
 - Cross-validation in questions, interviewees, checking interpretations
- Example study?
 - Herbsleb & Grinter: software integration

Example: Interviews Rich precise

- What contributions can it support?
 - Nature of problem, as perceived, current process
 - Examples, exceptions, incidents
- What are its limitations?
 - Information is processed, filtered by interviewees
 - May be inappropriately biased by questions
- What resources are needed?
 - Willing interviewees, correctly positioned
 - Ability to sample all relevant perspectives
 - Preparation, follow-up

Example: Repository Analysis, Rich Precise

- What is it?
 - Naturally occurring difference
 - Examination of digital traces in online repos
- What questions can it answer?
 - Exploratory
 - Base rate
 - Correlation
- What makes it rigorous?
 - Access to large-scale data
 - Quasi-groups available
- Example study?
 - Ray et al. A Large-Scale Study of Programming Languages and Code Quality in Github

Example: Repository Analysis, Rich Precise

- What contributions can it support?
 - Knowledge mining
 - Value of tool, method, process
- What are its limitations?
 - No cause-effect
 - Quasi-groups not random
- What resources does it take to do well?
 - Lot of data mining and stats
 - Contextualized information

Example: True Experiment Rich precise

- What is it?
 - Comparison that is engineered
 - Random assignment of values of independent vars
- What questions can it answer?
 - Cause and effect
 - Size of effect, interaction of factors
- What makes it rigorous?
 - Well designed experimental and control conditions
 - Attention to reliability, validity
- Example study?
 - Dekel et al: pushing directive information

Example: True Experiment Rich precise

- What contributions can it support?
 - Value of tool, method, process, training
 - Influence of context factors
- What are its limitations?
 - Must be able to isolate critical variables
 - Limited by ability to create situations that manipulate people
- What resources does it take to do it well?
 - Access to appropriate subjects
 - Statistics, measurement instruments



Experiments as Clinical Trials

Why would we expect it to be better?

Why do we need to know?

What will we do with the answer?

Is drug A better than drug B?

Better at doing what?

Better in what way?

Better in what situations?



Why would we
expect it to
be better?

You gotta have a theory!



Some Definitions

- A **model** is an abstract representation of a phenomenon or set of related phenomena
 - ↳ Some details included, others excluded
- A **theory** is a set of statements that explain a set of phenomena
 - ↳ Serves to explain and predict
 - ↳ Precisely defined terminology
 - ↳ Concepts, relationships, causal inferences
 - ↳ (operational definitions for theoretical terms)
- A **hypothesis** is a testable statement derived from a theory
 - ↳ A hypothesis is not a theory!
- In CS, we have mostly *folk theories*



The Role of Theory Building

→ Theories lie at the heart of what it means to do science.

↳ Production of generalizable knowledge

→ Theory provides orientation for data collection

↳ Cannot observe the world without a theoretical perspective

→ Theories allow us to compare similar work

↳ Theories include precise definition for the key terms

↳ Theories provide a rationale for which phenomena to measure

→ Theories support analytical generalization

↳ Provide a deeper understanding of our empirical results

↳ ...and hence how they apply more generally

↳ Much more powerful than statistical generalization